

Tunable magnetocrystalline anisotropy and valley polarization in an intrinsic ferromagnetic Janus $2H$ -VTeSe monolayer

Cunquan Li  and Yukai An ^{*}*Key Laboratory of Display Materials and Photoelectric Devices, Ministry of Education,**Tianjin Key Laboratory for Photoelectric Materials and Devices, National Demonstration Center for Experimental Function Materials Education, School of Material Science and Engineering, Tianjin University of Technology, Tianjin 300384, China*

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Two-dimensional Janus monolayers with specular asymmetry exhibit excellent physicochemical properties and are a good candidate for optoelectronic, valley electronic, and nanoelectronics devices. Based on the density functional theory, the Janus $2H$ -VTeSe monolayer is an intrinsic ferromagnetic semiconductor with an indirect band gap of 0.324 eV and exhibits a good thermodynamic and kinetic stability, in-plane magnetocrystal anisotropy, large spontaneous valley polarization of 155 meV, and high Curie temperature (T_c) of 380 K. The biaxial strain ($-6\% < \varepsilon < 6\%$) can effectively tune the Janus $2H$ -VTeSe monolayer from the bipolar magnetic semiconductor phase to half-semiconductor, spin gapless semiconductor, and half-metallic phases. Moreover, the magnetocrystal anisotropy energy (MAE) is modulated by the strain from 0.54 meV ($\varepsilon = -6\%$) to 1.32 meV ($\varepsilon = -2\%$), and the easy [100] and hard [001] magnetic axes could be switched from each other by charge carrier doping. The calculated valley optical response of the Janus $2H$ -VTeSe monolayer exhibits a valley-selective circular dichroism. Due to the broken inversion and time-reversal symmetry, the valley polarization and Berry curvature can be continuously tuned by applying biaxial strains (modulation range 24.2%), external electric field (modulation range 2%), and varying the magnetization angle (0 to 180 degrees). The Janus $2H$ -VTeSe monolayer possesses intrinsic ferromagnetic ordering, large spontaneous valley polarization, high T_c of 380 K, and considerable MAE of 1.15 meV, giving it a potential application in the two-dimensional spintronic devices.

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I. INTRODUCTION

Two-dimensional (2D) materials have triggered extensive research due to their amazing physical properties and promising applications in spintronics, optoelectronics, and valleytronics devices. Recently, various 2D monolayers with a hexagonal lattice structure, such as graphene [1], stanine [2], and MoS₂ [3], have offered the possibility of the practical realization of valleytronics. The valley, which labels the degenerate energy extreme of a valence (conduction) band at the k points in momentum space, has the negligible inter-valley scattering between two large separate k points [4] and exhibits great promise for encoding and manipulating information carriers in the next-generation information technology. As a prominent 2D system, transition metal dichalcogenides chlorides (TMDs) are named K -valley materials [5,6], which own two degenerate but unequal valley states at the K and K' points of the hexagonal first Brillouin zone. However, most TMDs do not exhibit spontaneous valley polarization due to their intrinsic time-reversal symmetry. Until now, some extrinsic methods have been proposed to induce valley polarization in TMDs, including optical pumping with circularly polarized light [7], external magnetic fields [8], magnetic element doping [9], and magnetic proximity effects [10]. However, these methods are not ideal in practical

applications [11]. Recently, new types of 2D ferrovalley materials, such as VX_2 ($X = S, Se, Te$) [12], $VAgP_2Se_6$ [13], $LaBr_2$ [14], NbX_2 ($X = S, Se$) [15], Nb_3I_8 [16], $TiVI_6$ monolayers [17], $MXenes$ [18], etc., have been proposed. They possess inherent spontaneous valley polarization introduced by intrinsic ferromagnetism and provide an opportunity to address these challenges of exogenous valleytronics materials. Unfortunately, the low experimental feasibility, small intrinsic band gap, and low Curie temperature (T_c) hinder the practical application of these 2D ferrovalley materials. Therefore, the exploration of ferrovalley materials with high T_c and large valley polarization is urgent for next generation electronics. On the other hand, both $2H$ -VSe₂ and $2H$ -VTe₂ monolayers are intrinsically ferromagnetic semiconductors [12] with high T_c above room temperature [19,20], indicating that the realization of the experimental synthesis of $2H$ -VSe₂ [21] may provide an exciting new platform for spintronics. Recently, the ordered Janus MoSSe monolayer has been successfully synthesized by a modified chemical vapor deposition method [22]. The lack of mirror and time-reversal symmetry in the Janus ferrovalley systems further gives birth to valley degrees of freedom. Therefore, a deeper understanding of the intrinsic physical properties of Janus ferrovalley materials is necessary. In this work, the stability, electronic properties, magnetic properties, and control of valley freedom for the Janus $2H$ -VTeSe monolayer are systematically investigated using first-principles calculations. The Janus $2H$ -VTeSe monolayer exhibits the instinctive in-plane ferromagnetic

^{*}ykan@tjut.edu.cn

semiconductor character with high T_c of 380 K and large spontaneous valley polarization of -155 meV. The biaxial strain and charge carrier doping can well modulate the electronic structures and magnetocrystal anisotropy (MA) of the Janus $2H$ -VTeSe monolayer due to the variation of V $3d$ -orbital occupancy. With the time-reversal symmetry and inversion symmetry breaking, the valley polarization and Berry curvature are continuously tuned by applying biaxial strain, external electric field, and the changing of magnetization direction. The Janus $2H$ -VTeSe monolayer with modifiable electronic structure and magnetic properties can be considered as potential candidates for low-dimensional spintronic devices.

II. COMPUTATIONAL METHODS

The first-principles calculations are performed using the plane-wave basis Vienna Ab initio Simulation Package (VASP) [23,24] based on density functional theory. The generalized gradient approximation (GGA) method with Perdew-Burke-Ernzerhof (PBE) is used for the calculation of the exchange-correlation function [25,26]. The rotationally invariant local spin-density approximation (LSDA)+ U method [27] is used to handle the strongly correlated corrections of the V $3d$ electron, and the on-site Coulomb interaction parameter U and exchange interaction parameter J are set at 2.00 and 0.87 eV, respectively [28]. The plane-wave cut-off energy is set to 500 eV and an 18-Å-thick vacuum layer is added along the z direction to avoid adjacent periodic interactions. The first Brillouin zone is sampled using a $15 \times 15 \times 1$ Γ -centered Monkhorst-Pack grid [29]. During structural relaxation, the geometry is completely relaxed until the energy converges to less than 1×10^{-6} eV and the Hellmann-Feynman force is less than 0.01 eV Å $^{-1}$. The phonon spectrum calculation with a $4 \times 4 \times 1$ supercell is performed using the PHONOPY code [30,31] based on density functional perturbation theory (DFPT). An *ab initio* molecular dynamics (AIMD) simulation is performed in a moles volume-temperature (NVT) ensemble with Nosé Hoover thermostat at 300 K [32]. The DFT-D3 method is considered for the van der Waals correction [33]. The spin-orbit coupling (SOC) effect is considered for the calculations of the electronic structure. The construction of maximally localized Wannier functions (MLWFs) [34] and the calculation of Berry curvature are realized by the WANNIER90 package [35]. The Hall conductivity is calculated using WANNIERTOOLS [36] based on the tight-binding (TB) model. A fine $30 \times 30 \times 1$ grid is used in the Wannier interpolation. The Monte Carlo simulation open-source project MCSOLVER [37,38] based on the Wolff algorithm of the classical Heisenberg model is used to estimate the T_c of the Janus $2H$ -VTeSe monolayer. The VASPBERRY package is used to estimate its optical properties [39].

III. RESULTS AND DISCUSSION

Figure 1(a) shows the top and side views of the Janus $2H$ -VTeSe monolayer with C_{3v} symmetry, which is constructed and confirmed by the material design software (Medea) [40]. After structural optimization, the corresponding lattice constant/total energy are 3.46 Å/ -16.59 eV, and the corre-

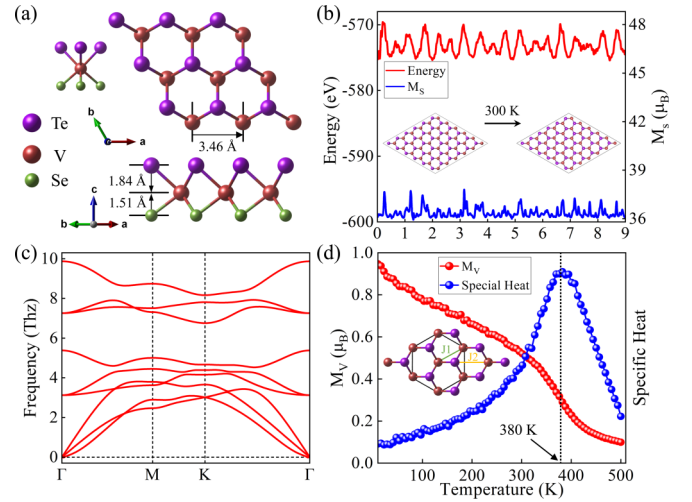


FIG. 1. (a) Top and side views of the Janus $2H$ -VTeSe monolayer. Purple, red, and green spheres represent Te, V, and Se atoms, respectively. (b) The evaluation of total free energy and magnetic moment during the 9 ps AIMD simulation for the $6 \times 6 \times 1$ Janus $2H$ -VTeSe monolayer. The inset is the initial (0 ps) and final (9 ps) supercells at 300 K. (c) The phonon spectrum of the Janus $2H$ -VTeSe monolayer. (d) Evolution of magnetic moment (red) and specific heat capacity (blue) for the Janus $2H$ -VTeSe monolayer in the $16 \times 16 \times 1$ superlattice based on the Monte Carlo simulation. The inset is the exchange path for the first (J1) and second (J2) nearest-neighbor V (red) pairs.

sponding V-Te and V-Se bond lengths are 2.72 Å and 2.51 Å, which are almost identical to those of $2H$ -VTe $_2$ and $2H$ -VSe $_2$ monolayers [41]. For a unit cell, the magnetic moments of V, Te, and Se atoms are 1.108, -0.124 , and -0.06 μ_B , respectively. The calculated average electrostatic potential along the z axis is rather asymmetric with a potential drop $\Delta\rho$ of 10.4 eV, as shown in Fig. S1 of the Supplemental Material (SM) [42]. The potential difference between the two sides of the V atom layer implies a change of work function, namely, $\Delta\phi = \phi_1 - \phi_2$. Obviously, the potential energy and work function on the Te side are smaller ($V_{Te} < V_{Se}$, $\phi_2 < \phi_1$) than those on the Se side, implying a larger electronegativity of the Te atom. The thermodynamic stability of the Janus $2H$ -VTeSe monolayer is further confirmed by performing AIMD calculations at 300 K. A $6 \times 6 \times 1$ supercell with a time step of 3 fs is used in the calculations. The fluctuations of total energy and magnetic moment as a function of time (9 ps) are shown in Fig. 1(b). The total energy of the supercell system oscillates around -573 eV and the total magnetic moment is about 36.0 μ_B . The snapshots of geometry also confirm that no significant structural damage is observed during the simulation. These results suggest that the Janus $2H$ -VTeSe monolayer is thermally stable with good magnetic order at room temperature. To further confirm the stability of the Janus $2H$ -VTeSe monolayer, the phonon band structure is calculated, as shown in Fig. 1(c). All the phonon frequencies are positive over the whole Brillouin zone, indicating the high dynamical stability for the Janus $2H$ -VTeSe monolayer. The spin densities of possible magnetic configurations including ferromagnetic (FM) and antiferromagnetic (AFM) orderings

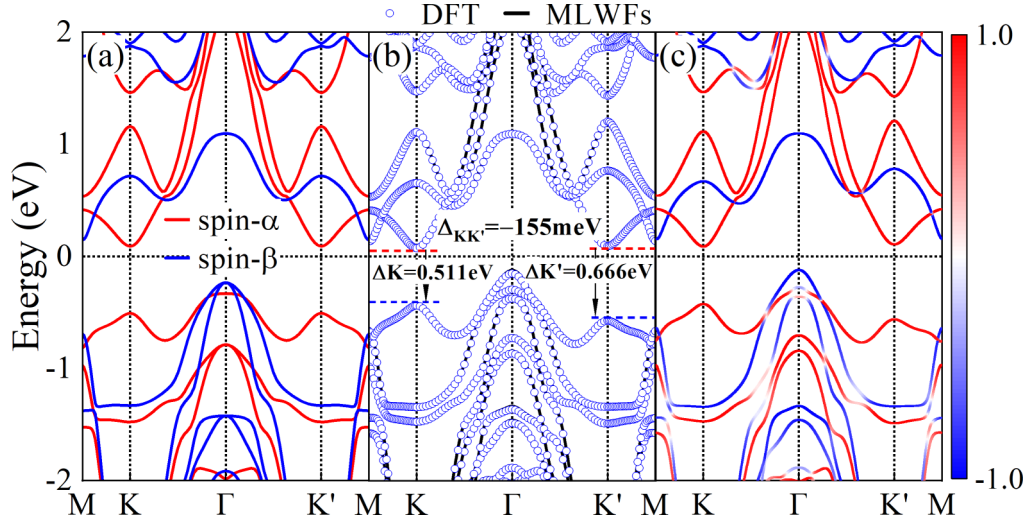


FIG. 2. (a) The spin polarization band structure of the Janus 2H-VTeSe monolayer. The red and blue lines indicate spin-up (α) and spin-down (β) electrons, respectively. (b) The band structure of the Janus 2H-VTeSe monolayer, considering SOC along the [001] direction in DFT+ U and MLWFs, respectively. (c) The band structure with the projection of spin operator $\hat{S}_z$ (color map). The red and blue colors indicate the spin-up and spin-down states, respectively. The Fermi level is set to zero.

are shown in Fig. S2 of the SM [42]. Clearly, the magnetic moment mainly localizes at the V atom for the Janus 2H-VTeSe monolayer. In a $2 \times 2 \times 1$ cell, four V atoms contribute a total magnetic moment of $4.43 \mu_B$ in the FM ordering. For the AFM ordering, the total magnetic moment equals $0 \mu_B$. The corresponding energy difference between the FM and AFM orderings can be defined as $\Delta E = E_{\text{FM}} - E_{\text{AFM}}$. The negative ΔE value (-0.476 eV) strongly suggests the intrinsic ferromagnetic ground state in the Janus 2H-VTeSe monolayer. 2D ferromagnetic monolayers with high T_c are highly desirable for practical applications in spintronic devices. Considering the disadvantage of mean-field theory, the fully anisotropic classical Heisenberg modes are used by the Monte Carlo simulation to determine the exact T_c for the Janus 2H-VTeSe monolayer. The spin Hamiltonian equation for the single-ion anisotropy can be written as

$$H = -J \sum_{\langle i, j \rangle} S_i \times S_j, \quad (1)$$

with

$$J = \frac{E_{\text{FM}} - E_{\text{AFM}}}{16S^2}, \quad (2)$$

where S_i is the spin operator, $\langle i, j \rangle$ denotes the nearest neighbors of the V atoms, J denotes the anisotropic spin-exchange interaction energy, and S is equal to 1. $J > 0$ favors ferromagnetic interactions. E_{FM} and E_{AFM} represent the total energy of FM and AFM orderings, respectively. The corresponding exchange parameter is evaluated with Eq. (2) and the calculated J value is 29.7 meV. To reduce the autocorrelation around T_c , the Wolff algorithm is implemented on a $16 \times 16 \times 1$ superlattice. Following the conventional work [28], only the nearest V-V interaction (J_1) is considered. Using this method, the T_c of the CrI₃ monolayer can be estimated to be 40 K, which is consistent with the experimental result of 45 K [43]. At each temperature, the spins of all magnetic points are flipped

randomly. 120 000 scans are used for fully thermalizing the system to equilibrium, and all statistics are obtained from the next 720 000 scans. As shown in Fig. 1(d), the corresponding T_c is predicted to be 380 K for the Janus 2H-VTeSe monolayer. T_c can also be estimated from the peak positions of specific heat C_v , showing the same results. Therefore, it can claim that the transition temperature from intrinsic ferromagnetic to paramagnetic ordering is about 380 K for the Janus 2H-VTeSe monolayer, which can be sustained at above room temperature.

The spin polarization band structure of the Janus 2H-VTeSe monolayer is shown in Fig. 2(a). One can see that the valence band maximum (VBM) located at the Γ point of the Brillouin zone is occupied by spin- β electrons, while the conduction band minimum (CBM) located at the K (K') point is contributed by spin- α electrons. Thus, the Janus 2H-VTeSe monolayer is a bipolar magnetic semiconductor (BMS) with an indirect energy gap of 0.324 eV, which lies between the 2H-VTe₂ (0.36 eV) and 2H-VSe₂ (0.28 eV) monolayers [12]. The band structure of the Janus 2H-VTeSe monolayer, considering the SOC effect, is shown in Fig. 2(b). The valley energies of the CBM at the K (73.8 meV) and K' (92.5 meV) points remain close, while the energy of the VBM at the K point (-436.9 meV) is remarkably higher than that at the K' point (-573.2 meV). Therefore, the valley gap ΔK and $\Delta K'$ at the K and K' points is 511 and 666 meV, respectively, resulting in a valley polarization ($\Delta_{KK'} = \Delta K - \Delta K'$) of -155 meV, which is larger than those in the Janus 2H-VSSe (85 meV) [28], 2H-VClBr (70 meV) [44], 2H-WSe₂ (1 meV) [45], and 2H-MoS₂ (3 meV) [46] monolayers and exhibits rather good development of 2D valleytronic devices. To ensure the calculation accuracy of the Wannier base functions, the tight-binding band structure using the MLWFs is calculated and also shown in Fig. 2(b). The band structure using MLWFs is highly consistent with the DFT results around the Fermi level, indicating that the obtained Wannier basis functions are sufficiently localized to ensure accuracy. Figure 2(c)

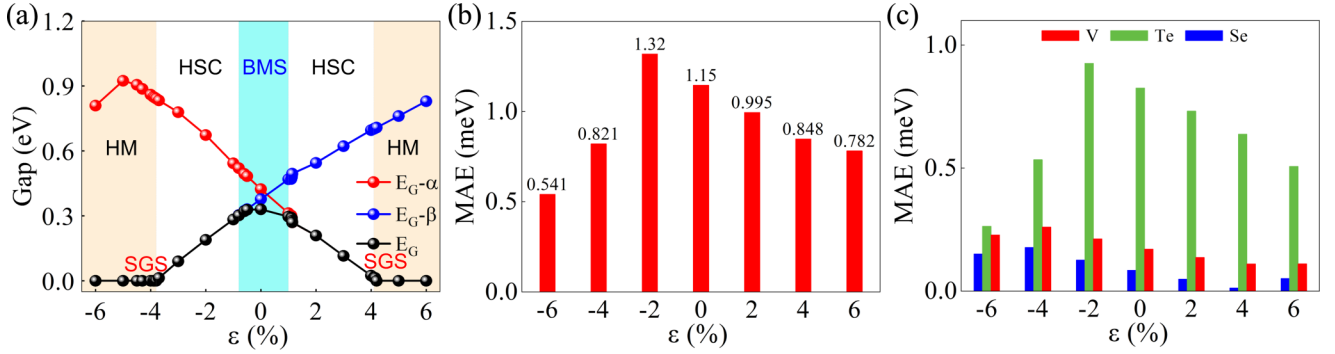


FIG. 3. (a) The dependence of band gap energy on biaxial strain. The red, blue, and black lines indicate $E_{G-\alpha}$ (spin up), $E_{G-\beta}$ (spin down), and E_G (total), respectively. The (b) total and (c) atomic-layer-resolved magnetocrystal anisotropy energies (MAEs) of the Janus $2H$ -VTeSe monolayer under various biaxial strains.

shows the band structure of the Janus $2H$ -VTeSe monolayer with the projection of spin operator $\hat{S}_z$, i.e., $\langle \psi_{nk} | \hat{S}_z | \psi_{nk} \rangle$, obtained from Wannier interpolation [47]. The spin-averaged moments at the K and K' valleys are equal, suggesting that the valley carriers in the VBM can only be distinguished by their valley index. In addition, the spontaneous valley polarization of the Janus $2H$ -VTeSe monolayer originates from the magnetic exchange interaction as well as the SOC effect, which is proportional to $\mu_L \cdot \mu_S$, where μ_L and μ_S represent the orbital and spin magnetic moments of the V atom, respectively. The calculated spin-average moments at the K and K' valleys of the VBM are $\hat{S}_x \approx \hat{S}_y \approx 0$ and $\hat{S}_z \approx 1$, where $\hat{S}$ is the spin operator. Thus, the spins at the K and K' valleys keep the same upward direction, resulting in generating an effective magnetic field B_{eff} and energy transfer of $B_{\text{eff}} \cdot \mu_L$. The orbital magnetic moment $\mu_L \approx 0\mu_B$ at the K and K' valleys of the CBM, and thus the corresponding energy transfer $B_{\text{eff}} \cdot \mu_L \approx 0$. In contrast, $\mu_L \approx \pm 2\mu_B$ at the K and K' valleys of the VBM, leading to an energy shift of $2\mu_B \cdot B_{\text{eff}}$. Therefore, the valley polarization $\Delta_{KK'} = 4\mu_B \cdot B_{\text{eff}}$. The ferromagnetic Janus $2H$ -VTeSe monolayer induces a very large valley polarization of 155 meV, implying that $B_{\text{eff}} \approx 693$ T. Therefore, it can consider that the intrinsic magnetic exchange interaction is much more effective in inducing the valley polarization. Figure S3 of the SM [42] shows the V d , Te p , and Se p orbital-resolved band structures, considering SOC. The K and K' valleys in the VBM are mainly dominated by the in-plane V d_{xy} and $d_{x^2-y^2}$ orbitals, whereas those in the CBM are dominated by the out-of-plane V d_{z^2} orbital. The contribution coming from the p orbitals of the Se/Te atoms is almost negligible. The 3D band structure over the first Brillouin zone (Fig. S4(a) of the SM [42]) as well as the 2D projected band structure of the VBM and CBM at the $k_x k_y$ plane for the Janus $2H$ -VTeSe monolayer considering SOC are shown in Figs. S4(b)–S4(d) of the SM [42]. The global minima (local maxima) of the energy at the K and K' points corresponding to the CBM (VBM) are called valleys. In the hexagonal crystal system, the six energetically degenerate valleys in the first Brillouin zone at the K and K' points are clearly shown, reflecting the obvious valley polarization character in the Janus $2H$ -VTeSe monolayer.

In-plane biaxial strain is often used to modulate the magnetic and electronic properties of 2D systems, which can be applied by the relation of $\varepsilon = [(a - a_0)/a_0] \times 100\%$, where

a and a_0 present the lattice constants of strained and unstrained Janus $2H$ -VTeSe monolayers. Figure 3(a) shows the dependence of the band gap on biaxial strain for the Janus $2H$ -VTeSe monolayer. It is obvious that the Janus $2H$ -VTeSe monolayer could be well tuned from the bipolar magnetic semiconductor (BMS) to half-semiconductor (HSC), spin gapless semiconductor (SGS), and half-metallic (HM) phases (the details of the phases are in Fig. S5 of the SM [42]) by applying biaxial strains from -6% to 6% (Fig. S6 of the SM [42]). The unstrained Janus $2H$ -VTeSe monolayer exhibits the BMS phase with the corresponding $E_{G-\alpha}$ (spin up) of 0.423 eV and $E_{G-\beta}$ (spin down) of 0.377 eV. When the small strain ($0\% < \varepsilon < 1.15\%$) is applied, the BMS phase is still maintained. In order to further understand the effect of strain on the band structures, the band-edge difference between spin- α and spin- β electrons at the Γ point is defined as $\Delta E_{\text{VBM}-\Gamma} = E_{\text{VBM}-\Gamma-\alpha} - E_{\text{VBM}-\Gamma-\beta}$. When the $\Delta E_{\text{VBM}-\Gamma}$ value is 0 eV ($\varepsilon = 1.15\%$), the transition from the BMS to HSC phases occurs. With the increase of tensile strain from 1.15% to 4.15%, the $\Delta E_{\text{VBM}-\Gamma}$ value continues to increase from 0 to 0.216 eV. Meanwhile, under the HSC phase, the $E_{G-\alpha}$ and E_G values show a monotonic linear decrease, which follows a relationship: $E_G (E_{G-\alpha}) = -9.15\varepsilon + 0.39$. At $\varepsilon = 4.15\%$, the corresponding $E_{\text{VBM}-\Gamma-\alpha}$ and $E_{\text{CBM}-K-\alpha}$ values are 0 eV, implying a transition from the HSC to SGS phases, namely, the spin- α (spin- β) channel exhibits the metallic (semiconducting) character. As the tensile strain continues to increase ($4.15\% < \varepsilon$), the system still maintains the HM phase. Under the compressive strain range, the band-edge difference between the spin- α electrons at the K point and the spin- β electrons at the M point can be defined as $\Delta E_{\text{CBM}-K-M} = E_{\text{CBM}-K-\alpha} - E_{\text{CBM}-M-\beta}$. When the ε increases to -0.8% ($\Delta E_{\text{CBM}-K-M} = 0$ eV), the CBM shifts to the M point from the K point, resulting the transition from the BMS to HSC phases. As the ε further increases from -1% to -3.8% , the $E_G (E_{G-\beta})$ decreases from 0.283 eV to 0 eV. In addition, the $E_G (E_{G-\beta})$ and ε values also follow the linear relationship: $E_G (E_{G-\beta}) = 9.97\varepsilon + 0.385$. At $\varepsilon = -3.8\%$, both the $E_{\text{VBM}-\Gamma-\beta}$ and $E_{\text{CBM}-M-\beta}$ values become 0 eV, indicating a transition from the HSC to SGS phases. Similar to the tensile strain case, the system maintains HM order with increasing of compressive strain ($\varepsilon < -3.8\%$). Obviously, biaxial strain can effectively modulate Janus $2H$ -VTeSe from the BMS to HSC, SGS, and HM phases.

As is well known, MAE can be defined as the energy difference along with the in-plane [100] and out-of-plane [001] magnetization direction, namely, $\text{MAE} = E_{001} - E_{100}$. The unstrained Janus 2H-VTeSe monolayer shows a large positive MAE value of 1.15 meV, indicating that the magnetic moment of the V atom is along the in-plane [100] direction, namely, in-plane magnetic anisotropy (IMA). The dependence of MAE on the polar angle (θ) is investigated for the Janus 2H-VTeSe monolayer. For the polar angle, the MAE can be defined as $\text{MAE} = E_{\text{FM}}^0 - E_{\text{FM}}^\alpha$, where $\alpha = 0$ denotes the out-of-plane aligned spins. The MAE shifts with the polar angle and its dependence on θ is described as

$$\text{MAE}(\theta) = \lambda \sin^2(\theta). \quad (3)$$

The λ is the single-site anisotropy constant and the positive value promotes in-plane magnetization. As shown in Fig. S7(a) of the SM [42], the polar angular dependence of MAE shows a twofold continuous spin symmetry and its pattern can be accurately described by the fitting function. The fit results obtained from the DFT data and Eq. (4) are plotted in Fig. S7(b) of the SM [42], showing a great agreement for $\lambda = 1145.93 \mu\text{eV}$. It is clear that MAE followed the sine function, suggesting that the MAE could be well modulated by the magnetization orientation. Figures 3(b) and 3(c) show the total and atomic-layer-resolved MAEs for the Janus 2H-VTeSe monolayers under various biaxial strains. The Janus 2H-VTeSe monolayer still keeps the IMA character under various strains from -6% to 6% . As the tensile strain increases from 0% to 6% , the MAE decreases monotonically. However, the MAE first increases to a maximum value of 1.32 meV at the compressive strain of -2% , and then gradually decreases with increasing the compressive strain to -6% . The atomic-layer-resolved MAE is shown in Fig. 3(c). The total MAE is mainly contributed by Te atoms, which is consistent with the fact that the nonmetallic atoms induce a stronger SOC effect. Interestingly, the V and Se/Te atoms always maintain the IMA character throughout the whole strain ranges.

To explore the origin and mechanism of MAE, the second-order perturbation theory is considered [48]. MAE can be defined as the orbital matrix element differences as follows:

$$\text{MAE} = \xi^2 \sum_{o,u} \frac{|\langle \psi_o | \hat{L}_z | \psi_u \rangle|^2 - |\langle \psi_o | \hat{L}_x | \psi_u \rangle|^2}{E_u - E_o}, \quad (4)$$

where ξ represents the SOC constants, $E_o(E_u)$ are the energies of occupied (nonoccupied) states, respectively, and $\hat{L}_z$ and $\hat{L}_x$ are the orbital angular momentum operators. Figures 4(a)–4(c) show the V d and Te/Se p orbital-resolved MAE for the Janus 2H-VTeSe monolayer at the strain of -6% , 0% , and 6% . It is obvious that the IMA behavior under various biaxial strains is mainly contributed by the matrix element differences (p_x/p_y orbitals) of the Te atom and ($d_{xy}/d_{x^2-y^2}$ orbitals) of the V atom. The matrix element difference of the Se (p_x/p_y orbitals) atom makes a rather small contribution to the IMA. On the other hand, clearly, the matrix element difference of the Te (p_z/p_y orbitals) atom shows a transition from the IMA to the perpendicular magnetic anisotropy (PMA) character [Fig. 4(b)]. Due to the presence of

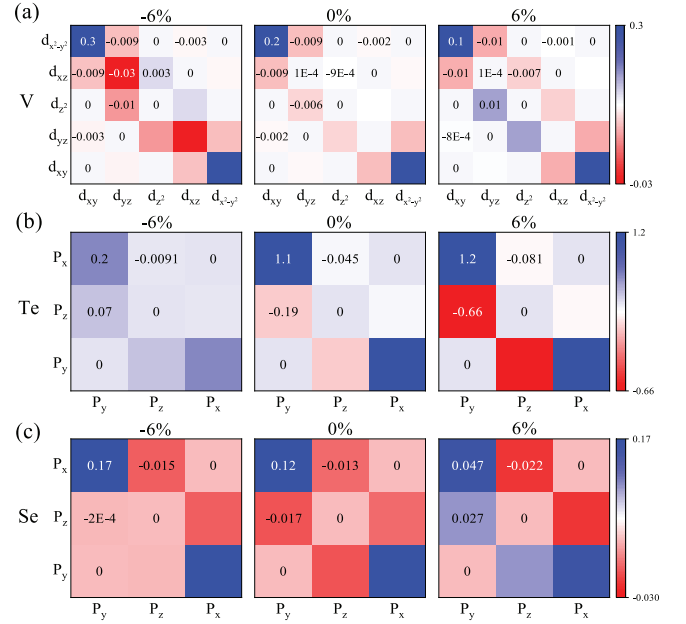


FIG. 4. (a) V d , (b) Te p , and (c) Se p orbital-resolved MAE for the Janus 2H-VTeSe monolayer at the strain of -6% , 0% , and 6% .

unpaired electrons in the V d orbital for the Janus 2H-VTeSe monolayer, it is expected that the MAE can also be tuned by charge carrier doping, which are shown in Figs. S8(a) and S8(b) of the SM [42]. The total MAE increases monotonically with increasing hole concentration, showing a PMA behavior, while an opposite behavior can be observed for the

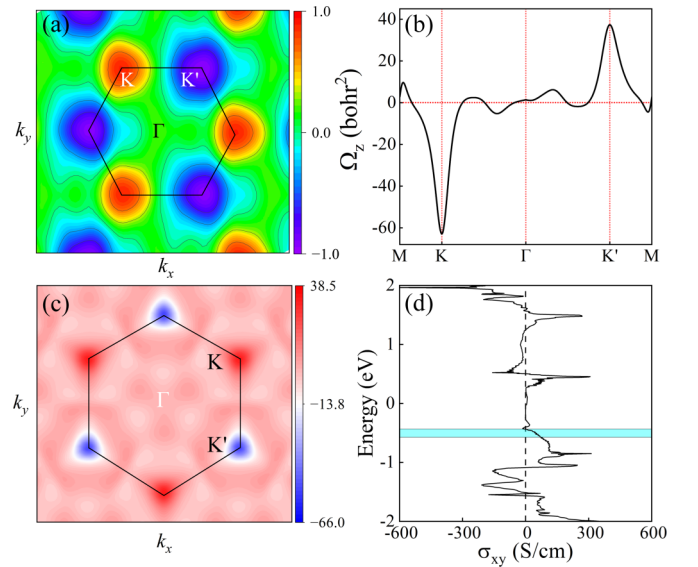


FIG. 5. (a) The circular polarization $\eta^{C,V}(k)$ of the optical transition between the VBM and CBM in the first Brillouin zone. The color scale indicates the values of $\eta^{C,V}(k)$ over the Brillouin zone. Calculated Berry curvature $\Omega_n(k)$ of the Janus 2H-VTeSe monolayer along (b) high-symmetry lines and (c) over the 2D Brillouin zone. (d) Calculated anomalous Hall conductivity of the Janus 2H-VTeSe monolayer. The two parallel lines indicate the two valley extremes. The magnetization vector along the out-of-plane [001] direction is considered for the above calculations.

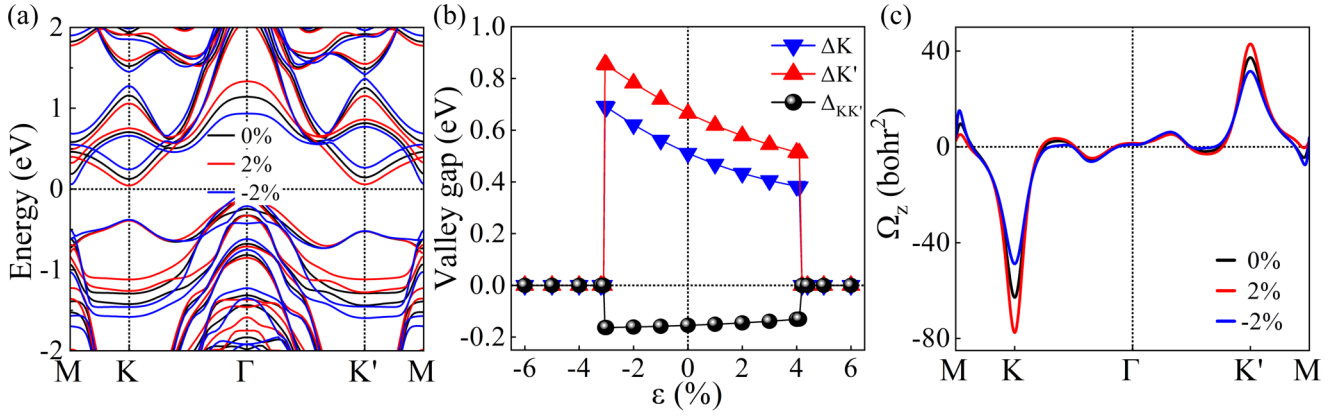


FIG. 6. The biaxial strain modulated (a) band structure, (b) valley gaps ΔK ($\Delta K'$) and valley polarization ($\Delta_{KK'}$), and (c) Berry curvature along high-symmetry lines of the Janus 2H-VTeSe monolayer.

electron doping, namely, the MAE decreases monotonically with electron doping, showing an IMA behavior.

For the nonmagnetic MoS₂ monolayer, there exists a valley-selective circular dichroism, which can selectively excite the valley carriers at the K or K' points with different circularly polarized light. Figure 5(a) shows the circular polarization $\eta^{C,V}(k)$ of the optical transition between the VBM and CBM in the first Brillouin zone for the Janus 2H-VTeSe monolayer. Generally, the $\eta^{C,V}(k)$ can be described as

$$\eta^{C,V}(k) = \frac{|P_+^{C,V}(k)|^2 - |P_-^{C,V}(k)|^2}{|P_+^{C,V}(k)|^2 + |P_-^{C,V}(k)|^2}. \quad (5)$$

The calculated interband transition matrix is $P_{\pm}^{C,V}(k) = \langle \psi_k^C | \hat{P}_{\pm} | \psi_k^V \rangle$, where $\hat{P}_{\pm} = (\hat{P}_x \pm i\hat{P}_y)/\sqrt{2}$. One can see that the $\eta^{C,V}(k)$ is nearly +1 and -1 at the K and K' points, respectively, which distinguishes the absorption of left and right circularly polarized (σ^+ and σ^-) light between the VBM and CBM. The left (σ^+) and right (σ^-) circularly polarized light can only excite the K and K' valley, respectively. Based on the above phenomena, circularly polarized light can be used to selectively excite the valley carriers. As the inversion and time-reversal symmetry is broken, the charge carriers acquire a valley-inverted Berry curvature $\Omega_n(k)$. According to the Kubo formula [49], the Berry curvature of the Janus 2H-VTeSe monolayer at the occupied state can be written as

$$\Omega_z(k) = \sum_n f_n \Omega_n(k) \quad (6)$$

and

$$\Omega_n(k) = - \sum_{m \neq n} \frac{2 \text{Im} \langle \psi_{nk} | v_x | \psi_{mk} \rangle \langle \psi_{mk} | v_y | \psi_{nk} \rangle}{[E_n(k) - E_m(k)]^2}, \quad (7)$$

where f_n is the Fermi-Dirac distribution function, and $|\psi_{nk}\rangle$ is the wave function of the Bloch states with the energy eigenvalue $E_n(k)$. The v_x and v_y are velocity operators along the x and y directions, respectively. Based on the above construction of the MLWFs, the Berry curvature of the Janus 2H-VTeSe monolayer is calculated by Eq. (7). Figure 5(b) shows $\Omega_n(k)$ of the Janus 2H-VTeSe monolayer along the high-symmetry lines. It can be seen that the Berry curvature $\Omega_n(k)$ exhibits two distinct unequal peaks at the K and K' valleys with opposite signs. Away from the K and K' valleys without inversion

symmetry, the $\Omega_n(k)$ decays rapidly and vanishes at the Γ point. At the same time, the distribution of $\Omega_n(k)$ in the 2D k plane is also calculated, as shown in Fig. 5(c), which clearly shows the C_3 symmetry of the system. And it is easy to observe that the K and K' valleys are distinguishable not only in terms of energy, but also in their Berry curvature. In fact, it can be found that the Berry curvature satisfies the relationship $\Omega_n(K)/\Omega_n(K') \approx -\Delta K^2/\Delta K'^2$. In the presence of an electric field E , the Berry curvature drives an anomalous transverse velocity of

$$v = -\frac{e}{\hbar} E \times \Omega_z(k), \quad (8)$$

which is proportionate to the valley-dependent Berry curvature. Consequently, this is an intrinsic contribution to the anomalous Hall effect. For the Janus 2H-VTeSe monolayer, the charge carriers in the K and K' valleys have opposite transverse velocities due to the unequal absolute values and opposite signs of Berry curvature. Thus, the Hall current can be measured, which provides a different way to realize information storage using the valley polarization. To demonstrate this, the anomalous Hall conductivity can be calculated by [8]

$$\sigma_{xy} = -\frac{e^2}{\hbar} \int \frac{d^2k}{(2\pi)^2} \Omega_n(k). \quad (9)$$

One can see that a nonzero σ_{xy} value can be generated as the Fermi level is located between two degenerate valley states in the VBM, as shown in Fig. 5(d).

Figure 6(a) shows the band structures of the Janus 2H-VTeSe monolayer at the strains of -2%, 0%, and 2%. Compared to the unstrained case, the tensile strain (2%) will move the bands down and the compressive strain (-2%) will press the bands up. Figure 6(b) shows the dependence of valley gaps ΔK ($\Delta K'$) and valley polarization ($\Delta_{KK'}$) on the various biaxial strains from -6% to 6%. Obviously, the ΔK ($\Delta K'$) values monotonously decrease with increasing the strains from -3.05% to 4.1%. Meantime, the average potential and the average potential drop $\Delta\rho$ of the Janus 2H-VTeSe monolayer under various strains is further calculated, as shown in Fig. S9(a) of the SM [42]. Obviously, the potential drop $\Delta\rho$ also monotonously decreases with strain and shows a similar behavior with valley polarization. The ΔK and $\Delta K'$ values exhibit a good quadratic dependence

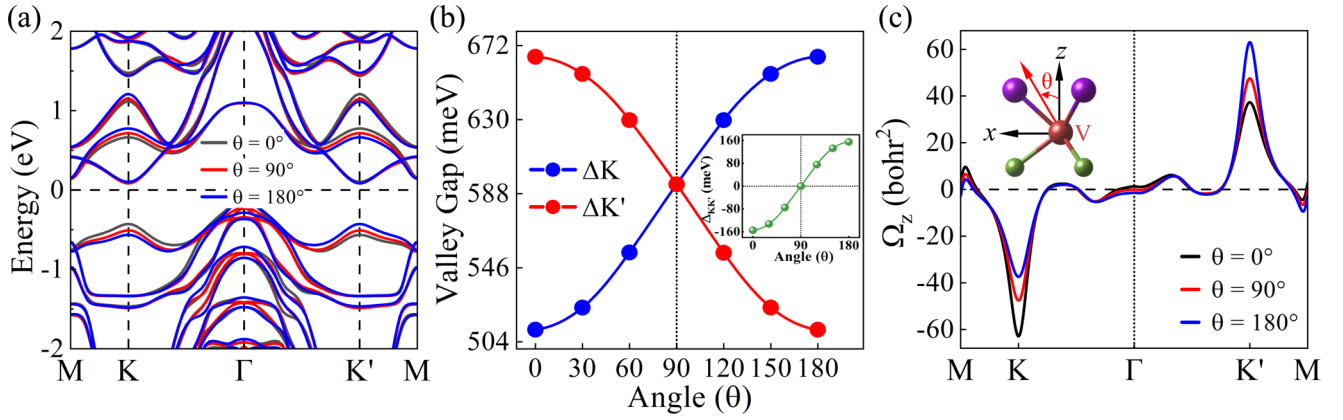


FIG. 7. The modulation of (a) band structures, (b) valley gaps ΔK ($\Delta K'$) and (the inset) valley polarization ($\Delta_{KK'}$), and (c) Berry curvature along high-symmetry lines of the Janus $2H$ -VTeSe monolayer by changing the magnetization angles θ between the magnetic moment and the z direction. The inset indicates the polar angle and the rotation plane of the magnetization direction.

on the biaxial strain, namely, $\Delta K = 38.6\varepsilon^2 - 4.74\varepsilon + 0.511$ and $\Delta K' = 34.3\varepsilon^2 - 5.15\varepsilon + 0.666$. The valley polarization ($\Delta_{KK'}$) reaches a maximum value of -163.3 meV at the strain of -3.05% and then decreases monotonously to a minimum value of -130.4 meV at the strain of 4.1% . Therefore, the valley polarization ($\Delta_{KK'}$) of the Janus $2H$ -VTeSe monolayer can be well modulated by biaxial strain. The Berry curvature of Janus $2H$ -VTeSe monolayers at the strains of -2% , 0% , and 2% is shown in Fig. 6(c). The maximum modulation of the Berry curvature can reach 24.2% at the strain of 2% and the tensile (compressive) strain tends to increase (decrease) the Berry curvature as well as their magnitude difference, $|\Omega_z(K) + \Omega_z(K')|$.

The polarity angle (θ) between the effective magnetic field and orbital magnetic moment also affects the valley polarization of the Janus $2H$ -VTeSe monolayer. The dependence of the band structures, valley gaps ΔK ($\Delta K'$), and valley polarization ($\Delta_{KK'}$) on the polarity angle (θ) is shown in Figs. 7(a) and 7(b). From 0 to 180 degrees, the ΔK ($\Delta K'$) value shows a monotonically increasing (decreasing) behavior at the K (K') valleys. Between 0 and 90 degrees, the valley polarization ($\Delta_{KK'}$) [the inset of Fig. 7(b)] decreases but remains negative. When rotating into the in-plane ($\theta = 90$ degree), the valley polarization vanishes because the symmetry is restored. With

further rotation from 90 to 180 degrees, the $\Delta_{KK'}$ becomes positive and increases continuously. In generally, the $\Delta_{KK'}$ can be modulated continuously from -155 meV to 155 meV. Figure 7(c) shows the Berry curvature along high-symmetry lines for the Janus $2H$ -VTeSe monolayer under various magnetization angles θ . The peak intensity of the Berry curvature at the K (K') point gradually decreases (increases) as the angle θ increases from 0 to 180 degrees. At the same time, the corresponding Berry curvature difference $|\Omega_z(K) - \Omega_z(K')|$ continues to increase monotonically, reaching a maximum value when the magnetization direction is perpendicular ($\theta = 180$ degrees) to the Janus $2H$ -VTeSe monolayer plane.

The X-V-Y type sandwich structure of the Janus $2H$ -VTeSe monolayer with broken mirror symmetry causes the net potential gradient perpendicular to the plane. Applying electric field E as a means of affecting the electron transfer can tune the potential of the Janus $2H$ -VTeSe monolayer and further control its electric and magnetic properties. As shown in Fig. S9(b) of the SM [42], the average potentials of the Janus $2H$ -VTeSe monolayer under external electric field E are calculated along the vertical z axis, and its positive direction is set from the Se atom to the Te atom, which causes the accumulation of more electrons near the Te and Se atoms. It is clear that the dipole moments (the inset of Fig. S9(b) of the

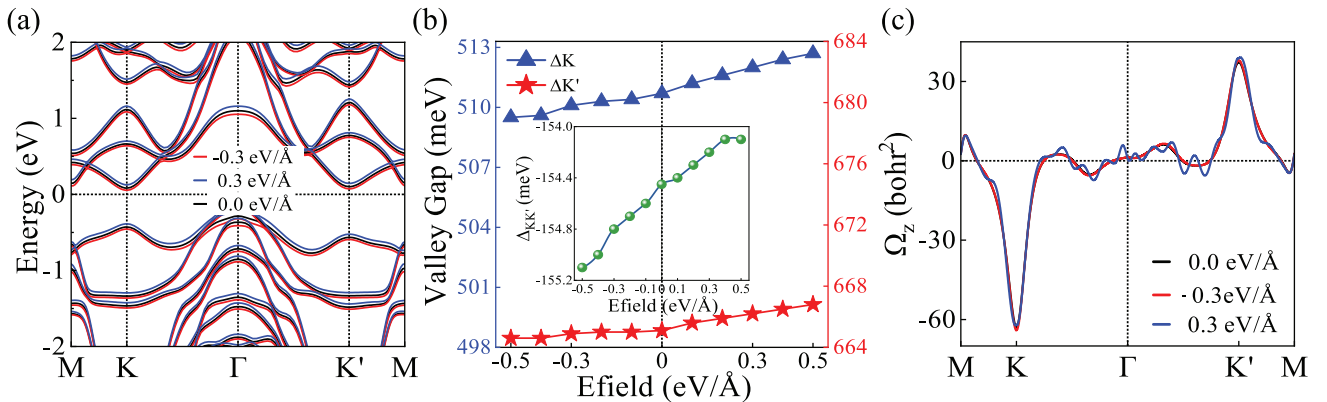


FIG. 8. The external electric-field dependence of (a) band structure, (b) valley gaps ΔK ($\Delta K'$) and (the inset) valley polarization ($\Delta_{KK'}$), and (c) Berry curvature along high-symmetry lines for the Janus $2H$ -VTeSe monolayer.

SM [42]) linearly increase from -0.96 debye to 0.66 debye under various electric fields from -0.5 eV/Å to $+0.5$ eV/Å. A similar behavior can also be observed for the potential drop $\Delta\rho$, implying that the charge transfer becomes stronger in the Janus $2H$ -VTeSe monolayer. As shown in Fig. 8(a), the electric field E can well adjust the band structures of the Janus $2H$ -VTeSe monolayer accompanied by the bands up at the positive electric field (0.3 eV/Å) and the bands down at the negative electric field (-0.3 eV/Å) with respect to the vacuum level, which is due to the enhancement of the dipolar field E and the overlap of V $3d$ and Te/Se $3p$ orbitals. The ΔK and $\Delta K'$ show a monotonic increase as the electric field E varies from -0.5 eV/Å to 0.5 eV/Å, and the $\Delta_{KK'}$ also exhibits the same behavior [the inset of Fig. 8(b)]. In addition, the $\Delta_{KK'}$ also increases (decreases) slightly as the dipole moments increase positively (negatively). The calculated Berry curvature $\Omega_n(k)$ modulated by an electric-field strength of ± 0.3 eV/Å is shown in Fig. 8(c). The effect of the electric field E is found to have a small effect on the Berry curvature $\Omega_n(k)$ at the K and K' valleys. These results reveal that the valley polarization and Berry curvature $\Omega_n(k)$ can be adjusted by changing the dipole moment under various electric field E , which can be explained by the regulation of the mirror-asymmetric charge distribution in the z direction.

IV. CONCLUSIONS

In summary, the electronic, magnetic, and valleytronic properties of the Janus $2H$ -VTeSe monolayer are

investigated in detail using first-principles calculations. The Janus $2H$ -VTeSe monolayer exhibits an intrinsically in-plane ferromagnetic semiconductor character with high T_c of 380 K and large spontaneous valley polarization of 155 meV. High stability is also well confirmed by the phonon structure and AIMD simulation. Under the in-plane biaxial strain, the Janus $2H$ -VTeSe monolayer undergoes a phase transition from the BMS to SGS, HSC, and HM phases. The MAE can be efficiently modulated by varying the magnetization angle, and by applying biaxial strain and charge carrier doping, especially since a transition from IMA to PMA character occurs under proper hole doping. The valley-selective circular dichroism persists and a scheme of valley-selective excitation by unpolarized light is thus proposed. The breaking of symmetry in the Janus $2H$ -VTeSe monolayer makes the valley freedom more tunable. The valley polarization can be continuously modulated from -163.3 meV to -130.4 meV under various biaxial strains, accompanied by the obvious variation of Berry curvature. The corresponding anomalous Hall conductivity (AHC) can be enhanced and the transverse Hall voltage can be effectively controlled.

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